Isaac Mubiru

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PROFESSIONAL SUMMARY

Full-stack product engineer with 6+ years of fully remote work who ships without a handed-down spec — **data pipelines, LLM tools and the UI**. Led a 4-developer team at CodeBits with no PM, working directly with NGO clients to decide what to build and shipping a backend, mobile app and USSD channel. At MTailor, shipped revenue features (**Express Shipping, +$40 per order**; a **video**-based 3D feature that increased conversion) and led a zero-downtime migration of 20+ apps. Builds with LLMs pragmatically: a production OpenAI + Vision assistant over WhatsApp and an **agentic pipeline** that verifies sources before any model call. Code: github.com/zac-09.

CORE COMPETENCIES

End-to-End Product Ownership | Data Pipelines (Pub/Sub, Kafka) | LLM Tools & Agentic Workflows | React UI Delivery | Node.js Backend Services & APIs | Video / Media Processing (ffmpeg) | Revenue & Conversion Features | Working Directly with Users

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote · e-commerce

- Built a customer-facing **3D visualisation feature** using video overlay and the **ffmpeg** library, increasing buyer conversion.
- Shipped **Express Shipping** end to end, generating an additional **$40 revenue per order**.
- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Built a real-time two-way **data pipeline** between MongoDB and Firestore using Node.js and Google Pub/Sub; migrated a Webflow website from a Parse backend to Firebase.
- Migrated file storage from Amazon S3 to Google Cloud Storage via a Python script; saved **$5,000/month** by moving all services off AWS.

MTailor

Mar 2022 – Jul 2022

Contractor · US Remote

- Built the migration script that read data from MongoDB, ran complex processing logic and wrote to Firestore; documented the new Firebase SDKs (Node.js and Python) and trained the Ops team.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote · fintech PWA

- Built ad-hoc jobs over **2M+ Firestore records** to keep real-time stock prices current from Morningstar APIs.
- Worked across the stack: responsive UI pages in **React** and Tailwind, and backend APIs in **TypeScript** on Firebase Cloud Functions and Express.js.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote · agency

- Full-stack delivery for demanding clients, reporting to the CTO: **React UIs from Figma designs** under tight timelines and Node.js backends for DeFi applications using Web3.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote · legal tech

- Led a team of 4 developers building complete products for NGOs: a case management system (FIDA Uganda) and a paralegal database (LASPNET Uganda).
- Built a USSD service for legal aid providers on a Node.js backend via gRPC; architected a microservices backend on **Apache Kafka, Docker and Kubernetes**.
- Developed a cross-platform mobile app in React Native and Expo (iOS & Android) with Firebase push notifications.

PROJECTS

Agentic Evaluation Pipeline agents

Extended an open-source agentic pipeline on Claude Code that scans job portals, verifies each source live in a headless browser before any model scoring, deduplicates inputs and generates tailored documents — 200+ evaluations run with verification-before-inference as the core rule.

Claude Code · Playwright · Node.js · YAML

Healthcare WhatsApp Assistant multimodal LLM

Conversational assistant that reads prescription photos and replies inside WhatsApp: OpenAI for dialogue, Google Vision for OCR, behind the WhatsApp Business API.

OpenAI API · Google Cloud Vision · WhatsApp Business API · Node.js

EDUCATION

B.Sc. Computer Engineering — Makerere University 2018 – 2022

Kampala, Uganda

SKILLS

Languages: JavaScript, TypeScript, Python, HTML, CSS, Go (intermediate), Rust (intermediate)

Frontend: React, Tailwind, PWA, React Native (intermediate)

Backend: Node.js, Express.js, REST APIs, gRPC, Microservices, Firebase Cloud Functions

AI: OpenAI API, Google Cloud Vision, Claude Code agentic workflows

Async & Data: Google Pub/Sub, Apache Kafka, NATS Streaming, MongoDB, Firestore

Cloud / Infra: GCP, Firebase, Google Cloud Storage, AWS (intermediate), Docker, Kubernetes, Linux, Git **Media:** ffmpeg (video overlay)

Spoken: English, Kiswahili